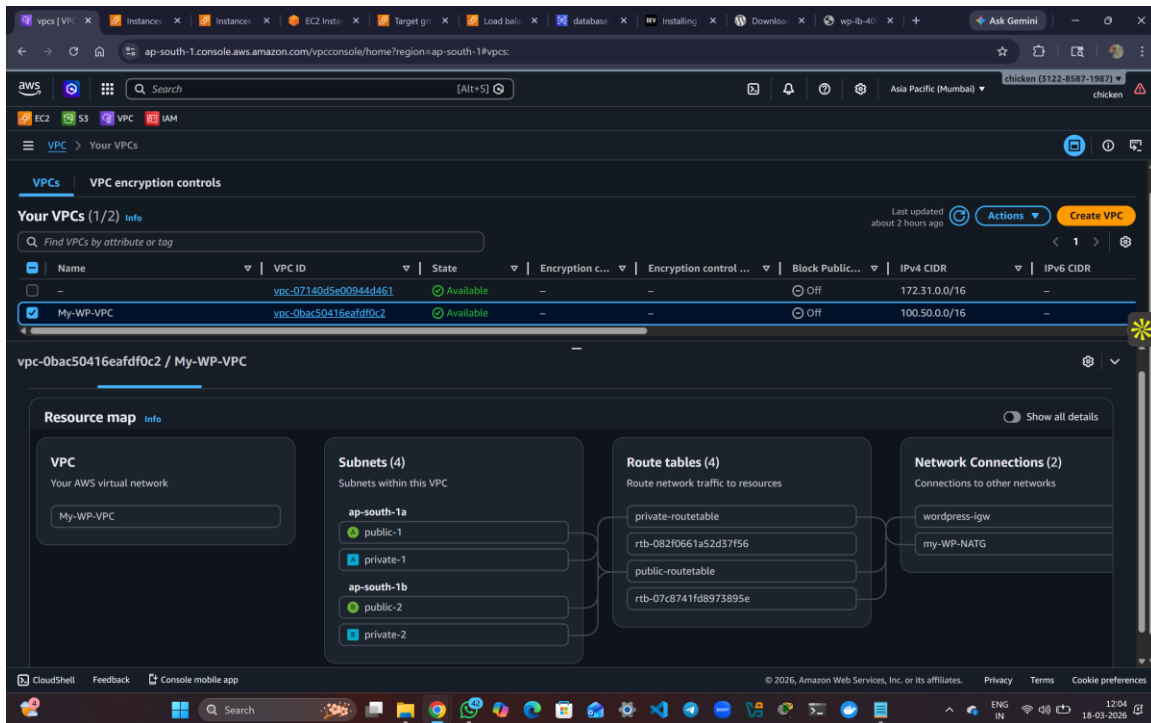


Created a custom virtual network with CIDR 100.50.0.0/16 to isolate resources.

Screenshot:



Step 2: Create Subnets

Description:

Created 2 public and 2 private subnets across multiple availability zones.

Step 3: Configure Internet Gateway

Description:

Attached Internet Gateway to allow public internet access.

Step 4: Configure NAT Gateway

Description:

Enabled private instances to access internet securely.

Step 5: Create Security Groups

Description:

Configured rules for bastion, web server, load balancer, and database.

Bastion-SG

- Inbound: SSH (22) → my IP
- Outbound: All traffic

ALB-SG

- Inbound: HTTP (80) → 0.0.0.0/0
- Outbound: All traffic

WebServer-SG

- Inbound:
 - HTTP (80) → ALB-SG
 - SSH (22) → Bastion-SG
- Outbound: All traffic

RDS-SG

- Inbound: MySQL (3306) → WebServer-SG
- Outbound: All traffic

Step 6: Launch Bastion Host

Description:

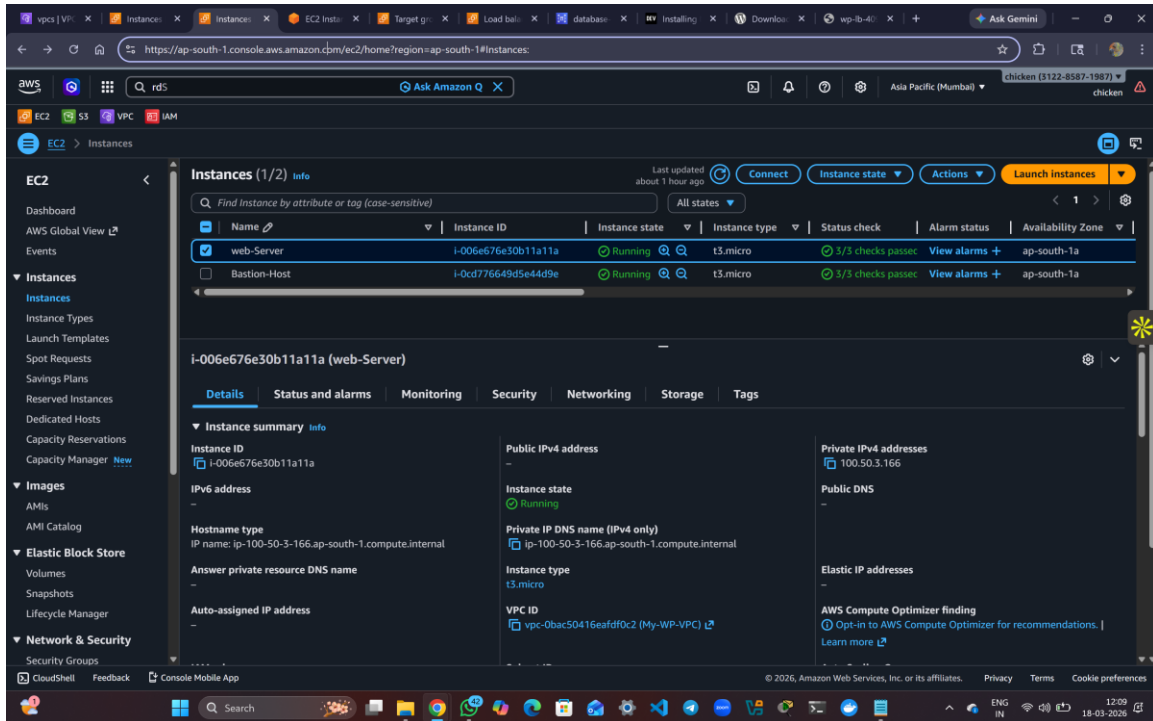
Deployed a public instance to securely access private servers.

Step 7: Launch Web Server

Description:

Created EC2 instance in private subnet for hosting WordPress.

Screenshot:



Step 8: Install Apache & PHP

Description:

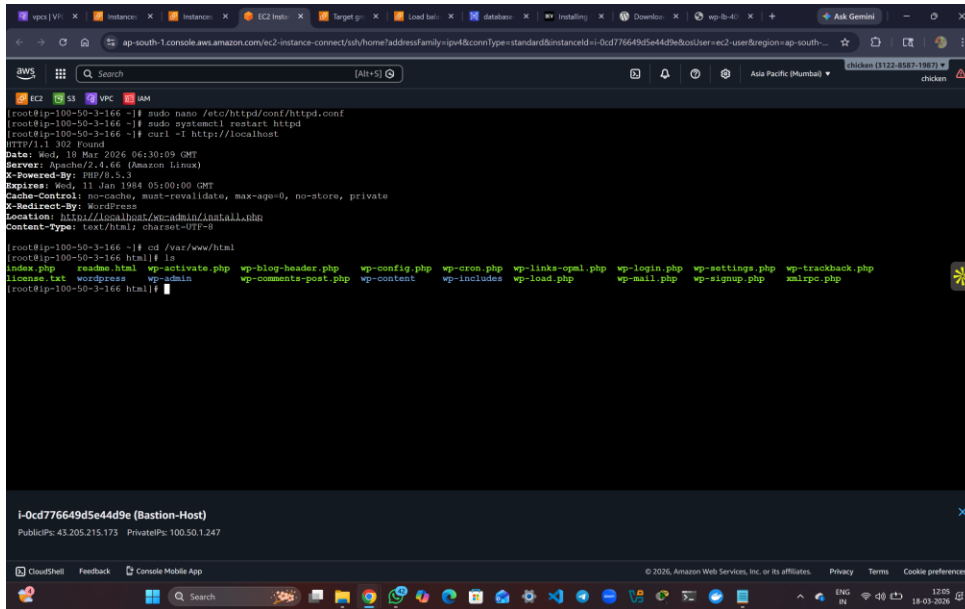
Installed web server and required PHP modules.

Step 9: Install WordPress

Description:

Downloaded and configured WordPress files.

Screenshot:

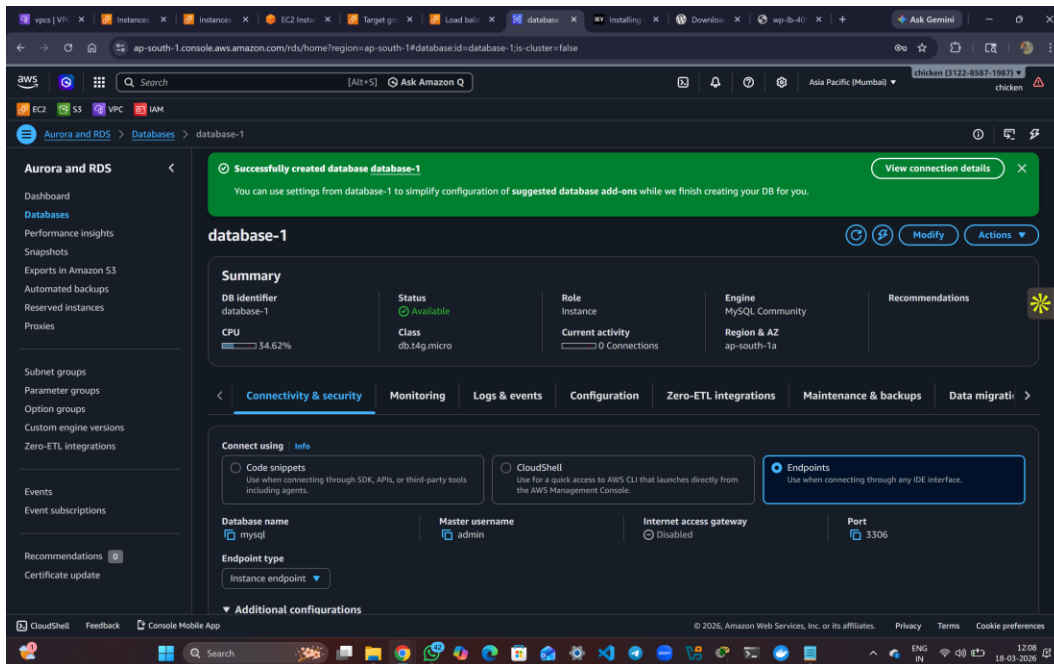


Step 10: Create RDS Database

Description:

Created MySQL database for WordPress.

Screenshot:



Step 11: Connect EC2 to RDS

Description:

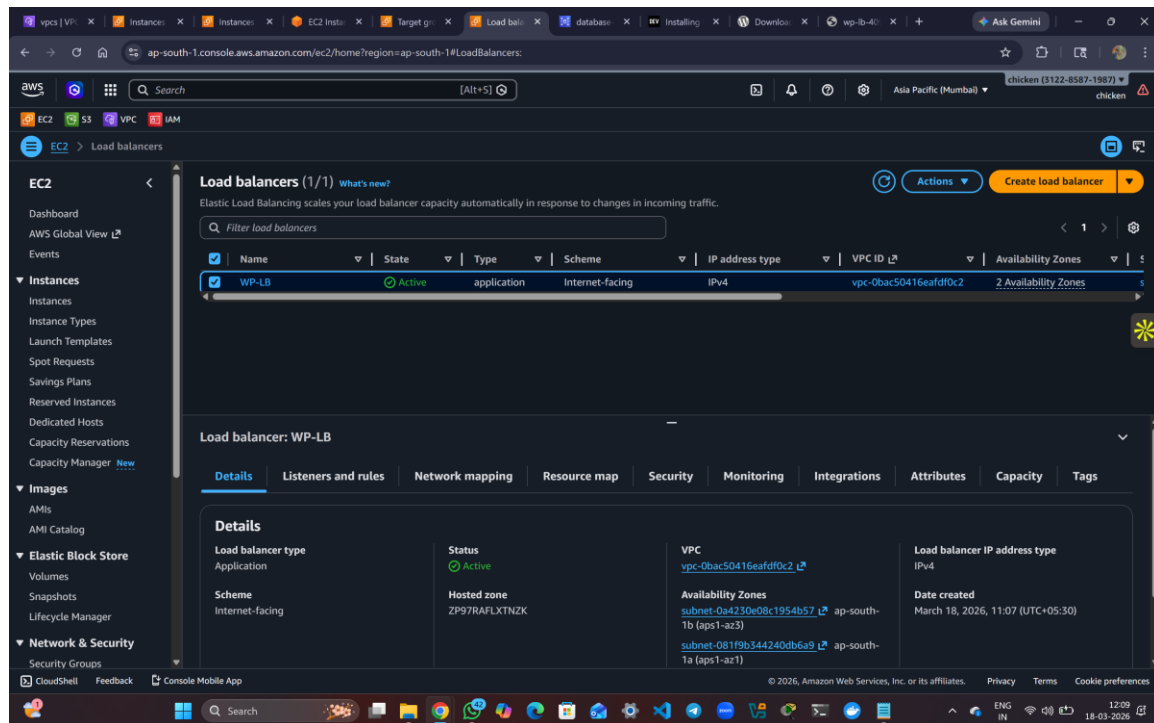
Configured database connection in wp-config.php.

Step 12: Create Load Balancer

Description:

Configured application load balancer for traffic distribution.

Screenshot:



Step 13: Create Target Group

Description:

Added EC2 instances and configured health checks.

Step 14: Fix Health Check Issues

Description:

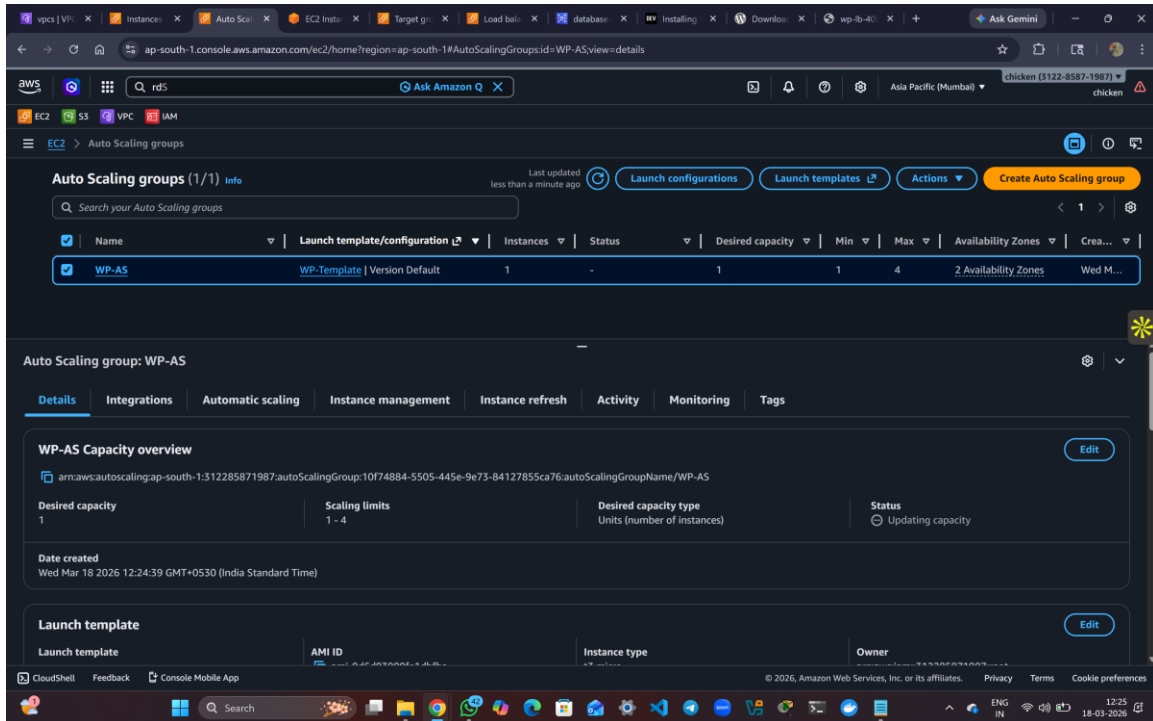
Resolved 403 errors for healthy status.

Step 15: Create Auto Scaling Group

Description:

Configured automatic scaling for high availability.

Screenshot:

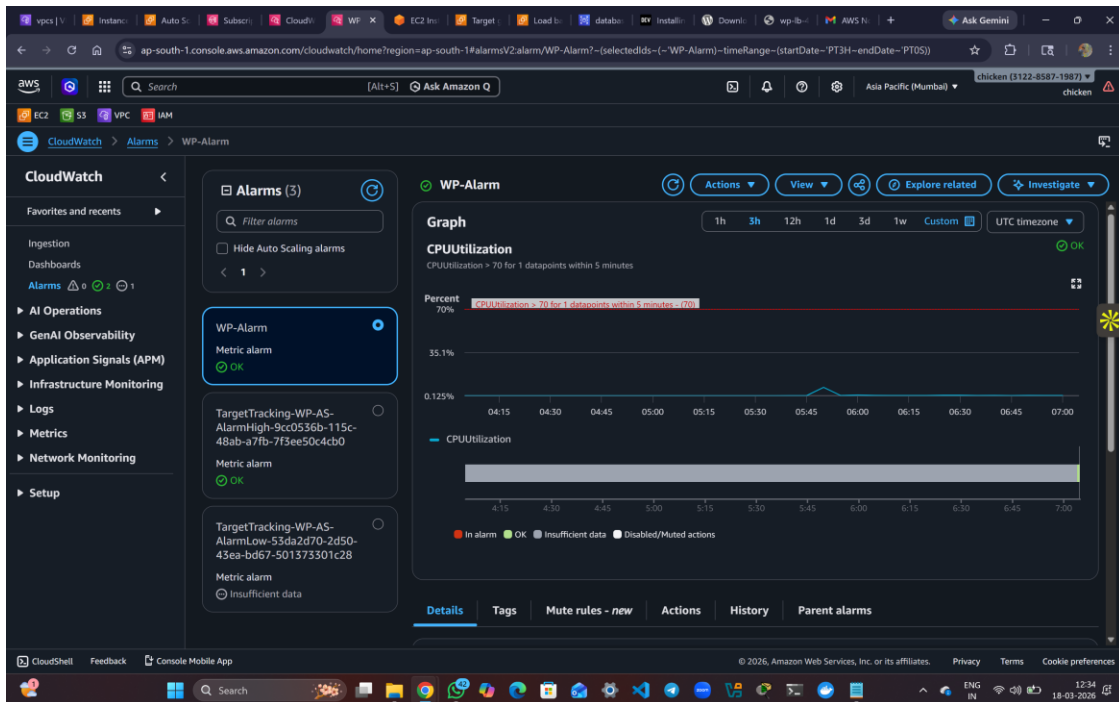


Step 16: Setup CloudWatch Alarm

Description:

Monitored CPU usage and system health.

Screenshot:

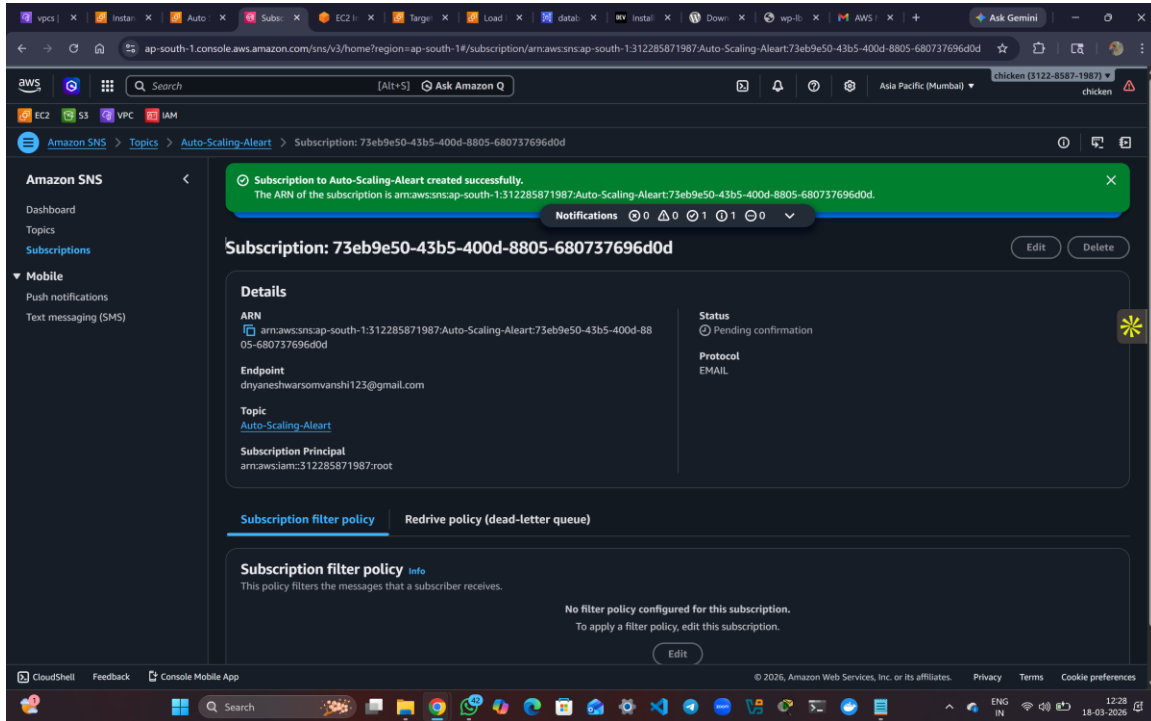


Step 17: Setup SNS Alerts

Description:

Configured email alerts for monitoring notifications

Screenshot:



Conclusion

This project demonstrates a real-world scalable and secure cloud deployment with monitoring and automation.